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A

MINOR PROJECT PROPOSAL

ON

”GAZE-DRIVEN VISUAL ATTENTION ANALYSIS USING WEBCAM-BASED EYE TRACKING”

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ABSTRACT

This project presents a webcam-based gaze-driven visual attention analysis system integrated with an embedded video player and image support to evaluate user engagement during multimodal media playback. The system estimates gaze direction using facial landmark detection and geometric projection methods, synchronizes gaze data with embedded video timestamps via platform-specific IFrame APIs or session timers for static images, and generates fixation-based heatmaps and engagement metrics. A key feature of the proposed framework is the integration of real-time Gaze Confidence Monitoring and post-session Accuracy Reporting. The system computes Focus Time and Distraction Time, providing quantitative measures of user attention by monitoring gaze within the media area. Evaluation is performed using a multi-stage approach involving calibration error, real-time signal quality, and engagement stability. The system aims to provide meaningful insights into viewer attention patterns while maintaining ethical data handling practices and offering a high-confidence analytical tool for research.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AOI	Area of Interest
API	Application Programming Interface
CNN	Convolutional Neural Network
CSV	Comma Separated Values
HCI	Human-Computer Interaction
I-DT	Identification by Dispersion-Threshold
KDE	Kernel Density Estimation
MCE	Mean Calibration Error
MPIIGaze	Max Planck Institute for Informatics Gaze Dataset
PnP	Perspective-n-Point
URL	Uniform Resource Locator

1. INTRODUCTION

1.1 Background

Visual attention is a fundamental component of cognitive processing, explicitly indicating where a user is directing their focus within a given visual environment. In fields such as Human-Computer Interaction (HCI), education, marketing, and usability research, understanding visual attention provides critical insights into user engagement, learning behaviors, and content effectiveness. Traditionally, visual analytics in digital media have been limited to high-level, aggregate metrics such as total views, click-through rates, and overall watch duration. While useful, these metrics fail to reveal the spatial distribution of attention—specifically, which regions of a screen or segments of a video are actively being observed versus those that are ignored.

To acquire granular visual attention data, researchers have historically relied on dedicated eye-tracking systems that utilize infrared (IR) illumination and pupil-center corneal reflection (PCCR) techniques [1]. While these hardware-based systems offer high spatial precision, they are prohibitively expensive, require controlled laboratory environments, and are difficult to deploy at scale. However, recent advancements in computer vision, particularly in facial landmark extraction and geometric head pose estimation, have catalyzed a shift toward appearance-based gaze estimation [2, 3]. These methods leverage standard, consumer-grade RGB webcams, eliminating the need for specialized hardware and significantly increasing accessibility.

Concurrently, the rapid evolution of web technologies has introduced robust browser-based capabilities, including WebRTC for real-time media capture and advanced JavaScript frameworks for client-side processing [4]. This technological confluence enables the development of web-based gaze tracking applications that can operate silently in the background while users interact with standard media interfaces. By integrating these tracking algorithms directly with platforms such as embedded IFrame Player APIs, it becomes possible to align real-time remote eye tracking [5] with dynamic video sequences and static image sessions. This paradigm shift democratizes visual behavior research, shifting it from specialized laboratories to everyday computing environments, and paves the way for scalable, synchronized attention analysis.

1.2 Problem Statement

Despite the widespread availability of digital video platforms and the growing necessity for detailed behavioral analytics, existing video metrics remain structurally superficial. They provide temporal engagement data, such as how long a video was played, but intrinsically fail to identify the specific visual regions within the frame that cap-

ture and sustain a user's attention. While high-precision, hardware-based eye-tracking systems excel at acquiring detailed spatial gaze data, their reliance on costly and specialized infrared equipment limits their accessibility, portability, and scalability for broad, general-purpose applications.

Conversely, modern software-only webcam trackers have emerged to bridge this accessibility gap, but they frequently operate as standalone tools. A major limitation of these software-based trackers is their inability to maintain tight, synchronous integration with dynamic, third-party media sources, such as embedded video players. When a user watches a dynamic video, their gaze coordinates are only meaningful if they are temporally locked to the exact frame of the video being displayed. If the video buffers, pauses, or is skipped forward by the user, standalone trackers lose contextual synchronization, rendering the collected spatial data invalid for determining specific content focus.

Furthermore, raw gaze coordinate data is inherently noisy and difficult to interpret without sophisticated post-processing. There is a distinct lack of hardware-independent systems that not only track gaze but also translate those raw coordinates into meaningful analytical insights—such as detecting stable fixations, distinguishing between focused and distracted states, and generating visual attention heatmaps overlaying the original media. Consequently, there is a critical need for an integrated, robust system capable of capturing webcam-based gaze data, dynamically synchronizing it with video playback states via platform APIs, and processing it into actionable, synchronized visual analytics without requiring specialized equipment or disrupting the natural user experience.

1.3 Objectives

- Develop an accessible, real-time webcam-based gaze tracking system using geometric estimations to determine visual attention.
- Establish seamless synchronization of gaze data with both dynamic embedded videos and static media content.
- Compute and visualize quantitative engagement metrics, including fixation heatmaps, attention timelines, and Focus/Distraction Time.
- Implement an end-to-end evaluation workflow featuring pre-session accuracy verification and real-time confidence monitoring.

1.4 Scope of the Project

The system is limited to single-user webcam-based tracking and is intended for academic and analytical purposes, not clinical-grade precision.

1.5 Significance and Limitation

The project provides a low-cost alternative for attention analysis that uniquely combines webcam-based gaze tracking with real-time embedded video synchronization to generate content-specific attention analytics without requiring specialized eye-tracking hardware. Unlike existing systems that either focus solely on gaze estimation or require expensive infrared equipment, this system delivers an end-to-end pipeline from passive gaze capture to interpretable attention visualization.

However, accuracy depends on lighting conditions, webcam quality, head movement, and user calibration. The system provides approximate gaze estimation suitable for academic and analytical purposes rather than clinical-grade precision.

2. LITERATURE REVIEW

Eye tracking research has significantly evolved over the past two decades. Early systems primarily relied on infrared (IR) hardware-based solutions for high-precision gaze estimation [1]. While these systems achieved strong spatial accuracy, they required specialized hardware, controlled environments, and high implementation cost.

With the advancement of computer vision and machine learning, appearance-based gaze estimation methods have emerged. These approaches leverage convolutional neural networks (CNNs) trained on large-scale datasets such as MPIIGaze to estimate gaze direction using standard RGB cameras [2]. Appearance-based methods reduce hardware dependency and enable gaze tracking using consumer-grade webcams.

Recent research has explored webcam-based gaze estimation, remote eye tracking, and real-time interaction techniques [5]. Studies have demonstrated that geometric head pose estimation combined with facial landmark detection can provide practical gaze approximation without requiring dedicated eye-tracking hardware [3]. Furthermore, web-based interaction frameworks have shown the feasibility of integrating gaze tracking into browser environments for human-computer interaction applications [4].

Several works have also investigated gaze-supported interaction in video-based environments, attention modeling, and behavioral analytics [6, 7]. These systems highlight the potential of gaze information for understanding user engagement, learning behavior, and visual attention patterns.

2.1 Existing Systems

Traditional eye-tracking systems utilize infrared illumination and corneal reflection techniques to achieve high angular precision [1]. These systems are widely used in cognitive research, usability studies, and medical diagnostics but are expensive and not easily deployable for general-purpose applications.

Appearance-based methods, such as those trained on MPIIGaze [2], use deep learning models to directly estimate gaze direction from eye images. While these approaches improve flexibility and reduce hardware requirements, they often focus primarily on improving gaze prediction accuracy rather than integrating gaze information into higher-level behavioral analysis frameworks.

Recent web-based and interactive systems demonstrate the feasibility of webcam-based gaze tracking in real-time environments [8, 9]. However, many existing solutions emphasize gaze estimation performance and do not provide synchronized video attention visualization, fixation-level interpretation, or engagement metric computation within a unified system.

2.2 Research Gaps

Despite substantial advancements in gaze estimation techniques, several gaps remain:

- Many high-accuracy systems require specialized infrared hardware.
- Deep learning-based models require large labeled datasets and significant computational resources.
- Limited systems integrate gaze estimation with synchronized video playback for attention-aware analysis.
- Few approaches provide interpretable outputs such as fixation detection, temporal engagement trends, and heatmap-based visualization using only consumer-grade webcams.

Therefore, there is a need for a lightweight, web-based gaze-driven attention analysis system that integrates real-time webcam tracking, synchronized video interaction (including uploaded and embedded sources), fixation detection, and interpretable visualization within a single unified framework.

The proposed system addresses this gap by combining geometric gaze estimation, head pose estimation, calibration-based correction, video timestamp synchronization, fixation detection, and behavioral attention analytics in a browser-accessible environment.

3. RELATED THEORY

This chapter presents the theoretical foundations underlying the proposed gaze-driven visual attention analysis system. The system integrates principles from computer vision, geometric modeling, human visual behavior, and statistical density estimation.

3.1 Human Visual Attention Theory

Human visual attention refers to the cognitive process of selectively concentrating on specific visual stimuli while ignoring others. Eye movements are typically categorized into:

- **Fixations:** Periods where the eye remains relatively stable and visual information is acquired.
- **Saccades:** Rapid eye movements between fixations.

Research in Human-Computer Interaction (HCI) suggests that fixation duration and frequency correlate strongly with user interest and cognitive processing load. Therefore, fixation detection forms the basis of attention analysis in this system.

3.2 Camera Geometry and Pinhole Camera Model

The system assumes a pinhole camera model to relate 3D world coordinates to 2D image coordinates.

The projection from a 3D point (X, Y, Z) to image coordinates (x, y) is defined as:

$$x = \frac{fX}{Z}, \quad y = \frac{fY}{Z}$$

where:

- f is the focal length of the camera,
- Z represents depth,
- (X, Y, Z) are 3D coordinates in camera space.

This model is fundamental for head pose estimation and gaze projection.

3.3 Facial Landmark Detection

Facial landmark detection identifies key feature points on the face such as eye corners, iris center, nose tip, and mouth corners. These landmarks provide geometric references required for head pose estimation and gaze computation.

Pre-trained deep learning models (e.g., regression-based or CNN-based landmark detectors) estimate landmark coordinates in real time from RGB images [10].

3.4 Head Pose Estimation

Head pose estimation determines the orientation of the head relative to the camera using rotation angles:

- Yaw (horizontal rotation)
- Pitch (vertical rotation)
- Roll (tilt)

The Perspective-n-Point (PnP) algorithm is used to compute head pose by solving the transformation between known 3D facial reference points and their corresponding 2D image projections [3].

The transformation is expressed as:

$$s \cdot \mathbf{p}_{2D} = \mathbf{K}[\mathbf{R}|\mathbf{t}]\mathbf{P}_{3D}$$

where:

- $\mathbf{K}$ is the camera intrinsic matrix,
- $\mathbf{R}$ is the rotation matrix,
- $\mathbf{t}$ is the translation vector,
- $\mathbf{P}_{3D}$ is a 3D reference point,
- $\mathbf{p}_{2D}$ is its 2D image projection.

3.5 Geometric Gaze Estimation

Geometric gaze estimation computes a gaze vector based on eye center position and head orientation.

The gaze direction vector $\mathbf{g}$ can be approximated as:

$$\mathbf{g} = \mathbf{R} \cdot \mathbf{v}_{eye}$$

where:

- $\mathbf{R}$ is the head rotation matrix,
- $\mathbf{v}_{eye}$ is the eye direction vector in local coordinates.

The resulting 3D gaze vector is projected onto the 2D screen plane after calibration.

3.6 Screen Coordinate Mapping and Calibration

Due to systematic bias and individual anatomical differences, a calibration phase is required.

Calibration involves:

- Displaying predefined target points on screen.
- Recording predicted gaze coordinates.
- Computing a transformation function (e.g., affine mapping).

The affine transformation is given by:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} + \mathbf{b}$$

where **A** represents scaling/rotation and **b** represents translation.

3.7 Fixation Detection Theory (I-DT Algorithm)

The Dispersion-Threshold Identification (I-DT) algorithm detects fixations by clustering gaze points within a spatial threshold over a minimum time window.

Given a sequence of gaze points, dispersion is computed as:

$$D = (x_{max} - x_{min}) + (y_{max} - y_{min})$$

If dispersion D remains below a predefined threshold for a minimum duration (e.g., 100–200 ms), the cluster is classified as a fixation [11].

3.8 Kernel Density Estimation for Heatmap Generation

Kernel Density Estimation (KDE) is used to generate smooth heatmaps from discrete fixation points.

The density function is defined as:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

where:

- n is the number of gaze points,
- h is the bandwidth parameter,
- K is the kernel function (typically Gaussian).

KDE transforms sparse gaze data into continuous visual attention maps.

3.9 Engagement Metrics Theory

User engagement is quantified using temporal and spatial attention metrics:

- **Focus Time (T_{focus}):** The cumulative duration during a session where the user's estimated gaze is mapped within the boundaries of the Video Area of Interest (AOI).
- **Distraction Time ($T_{distract}$):** The cumulative duration where the user is present but looking outside the video boundaries, or when the face is not detected (suggesting the user has looked away).
- **Focus Score:** The ratio of focus time to total session time, expressed as a percentage:
$$Focus\ Score = \left(\frac{T_{focus}}{T_{total}}\right) \times 100\%$$
- **Total Fixation Duration:** Accumulated time of all detected fixations.
- **Attention percentage per AOI:** Distribution of focus across different video regions.

These metrics allow for an objective evaluation of visual attention patterns across multimodal media content.

3.10 Gaze Confidence Theory

Gaze confidence refers to the estimated reliability of a gaze prediction at a specific moment in time. Unlike accuracy, which is measured against a ground truth, confidence is estimated based on real-time signal quality.

The Gaze Confidence Score (S_{conf}) is modeled as a product of face detection probability and landmark stability:

$$S_{conf} = P_{face} \times \left(1 - \frac{\sigma_{landmarks}}{\text{threshold}}\right) \tag{3.1}$$

where:

- P_{face} is the detection confidence score from the facial landmark model (ranging from 0 to 1).
- $\sigma_{landmarks}$ represents the spatial variance or "jitter" of key eye landmarks over a short window.

A high S_{conf} indicates optimal tracking conditions, while a low score suggests poor lighting, extreme head angles, or occlusion.

4. FEASIBILITY STUDY

The feasibility of the proposed system is evaluated across five key dimensions: technical, economic, operational, legal, and scheduling feasibility.

4.1 Technical Feasibility

The proposed system is technically feasible as it relies on established computer vision libraries such as OpenCV [12] and MediaPipe [10] for facial landmark detection and estimation of head pose.

Video playback and synchronization are implemented using standard web technologies, including HTML5 video elements and embedded video APIs.

All required algorithms—PnP for head pose estimation, geometric gaze mapping, I-DT fixation detection, and Kernel Density Estimation for heatmap generation—are well-documented and implementable using available development tools.

The system can run on standard consumer hardware without requiring specialized eye-tracking equipment.

4.2 Economic Feasibility

The project requires minimal financial investment. A standard RGB webcam and personal computer are sufficient for implementation.

All software tools and frameworks used are open-source or freely available for academic purposes. Therefore, the overall development cost is negligible.

The potential academic and research value significantly outweighs the minimal resource requirements.

4.3 Operational Feasibility

The system is designed for ease of use with a simple calibration procedure and an intuitive web-based interface.

Supports both locally uploaded videos and embedded platforms, increasing flexibility in real-world academic and research environments.

The application operates locally, ensuring smooth execution and reduced dependency on external infrastructure.

4.4 Legal and Ethical Feasibility

The proposed system involves passive tracking of user eye movements, which requires strict adherence to ethical standards. To ensure privacy and compliance:

- **Data Privacy:** All gaze estimation, fixation detection, and attention analysis are

performed locally on the user's device. No gaze data, video timestamps, or facial images are transmitted to external servers.

- **Landmark-Only Processing:** The system processes facial landmarks solely for gaze estimation. It does not perform facial recognition, identity verification, or store biometric profiles.
- **Informed Consent:** Explicit user consent is required before accessing the webcam. Users are clearly informed that their eye movements are processed strictly for attention analysis and academic research.

By processing all data locally and requiring explicit consent, the system remains compliant with both academic ethical guidelines and data protection standards.

4.5 Scheduling Feasibility

The project is modular in design, allowing parallel development of components such as gaze tracking, video synchronization, and attention analysis.

Since the system leverages pretrained models and established algorithms, development complexity is reduced.

Given the defined scope and available tools, the project is achievable within the allocated minor project timeline.

5. METHODOLOGY

5.1 System Architecture and End-to-End Workflow

Figure 5.1 illustrates the overall processing pipeline and system architecture of the proposed gaze-based attention analysis system. The architecture is designed to support the entire end-to-end user workflow, from application initiation to post-session reporting, integrating real-time system tracking with continuous confidence evaluation.

The comprehensive user journey and associated architectural components unfold across four distinct phases:

1. Initialization and Media Load: The user opens the web application and inputs a desired video URL or selects an image. The **Media Player Module** initializes the embedded content (utilizing IFrame APIs for videos) and prepares the interface for seamless playback without disruptive visual tracking indicators.

2. Calibration (Accuracy Gate): Prior to media playback, the system invokes the **Calibration Module**. The user visually follows 5 to 9 predefined target dots on the screen. The system captures the **Webcam Input**, performs **Face and Landmark Detection**, and computes a **Mean Calibration Error (MCE)**. This acts as an initial "Accuracy Gate." If the accuracy meets a required threshold, the system proceeds; otherwise, it prompts a recalibration to guarantee data validity.

3. Execution (Real-Time Confidence Monitoring): As the media plays naturally, the **Gaze Estimation Module** operates silently in the background. It utilizes geometric projection to map eye landmarks to screen coordinates. Crucially, during this phase, the system maintains a **Gaze Confidence Score**, evaluating landmark stability and face detection probability in real-time. The **Media Synchronization Module** simultaneously tags each gaze coordinate with the precise video timestamp, and **Data Logging** securely stores the synchronized dataset locally.

4. Post-Processing and Reporting: Upon the conclusion of the media session, the system transitions into an analysis state. The **Attention Analysis Module** processes the logged data, utilizing the I-DT algorithm for fixation detection. Finally, the **Visualization Module** renders a consolidated **Session Report Dashboard**. This dashboard overlays KDE-based **Attention Heatmaps** onto the media, charts temporal engagement metrics (Focus vs. Distraction Time), and presents the final aggregate Accuracy and Confidence scores, affirming the reliability of the collected visual insights.

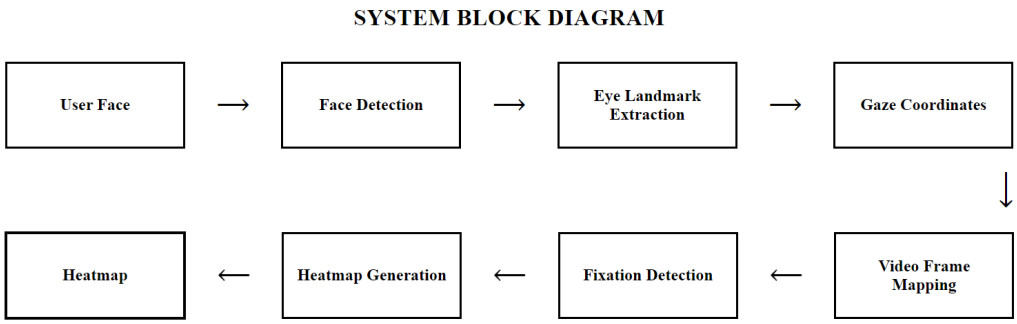

Figure 5.1: System block diagram

5.2 Work Flow of Proposed System

Figure 5.2 illustrates the detailed processing flow of the proposed system.

Unlike the system block diagram, which provides a high-level overview of major processing stages, the workflow diagram traces the step-by-step progression of data through the pipeline. Beginning from raw webcam frame capture, the data passes through face detection, landmark extraction, head pose estimation, and geometric gaze vector computation. The estimated gaze coordinates are then refined through calibration-based screen mapping before being synchronized with the current video timestamp. The synchronized gaze data undergoes fixation detection using the I-DT algorithm, followed by AOI classification and heatmap generation. Finally, engagement metrics are computed and visualized as attention overlays and temporal profiles.

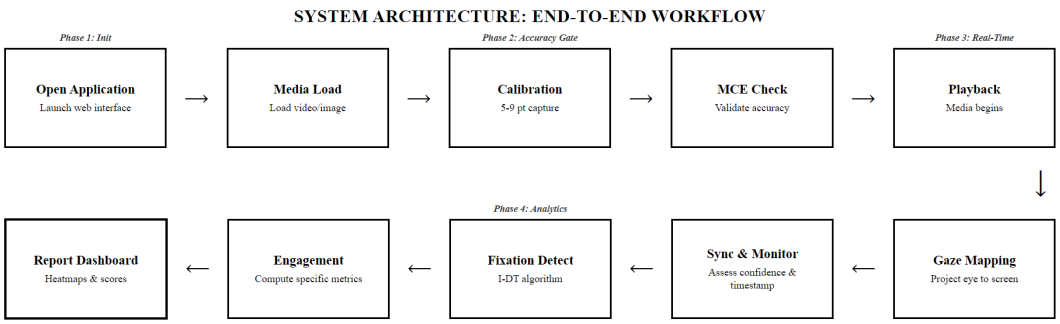

Figure 5.2: Work Flow

5.3 Performance Parameters

The following parameters are used to evaluate the performance of the proposed system:

- **Focus Score:** Percentage of total session time spent looking at the video content.
- **Distraction Index:** Cumulative duration of distracted states per session.

- **Gaze Confidence Score:** Real-time probability of tracking reliability based on facial landmark stability.
- **Calibration Error:** Mean Euclidean pixel distance between predicted gaze coordinates and known calibration target positions.
- **Region-Based Accuracy:** Percentage of gaze predictions that correctly fall within the intended screen region.
- **Fixation Stability:** Variance of gaze points within detected fixation clusters.
- **System Latency:** Average time delay between frame capture and gaze coordinate output.

The figure represents the detailed operational flowchart of the proposed system, illustrating the complete processing pipeline including all decision points, feedback loops, and the transition from real-time tracking to post-session analysis.

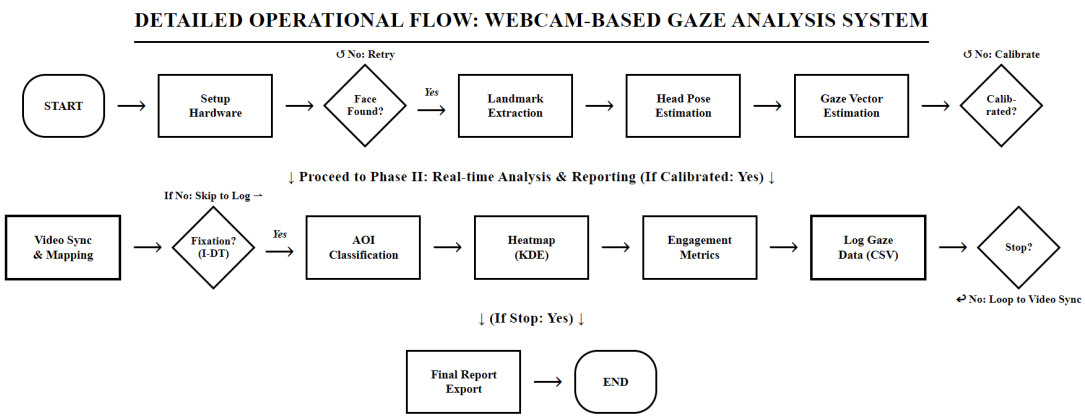

Figure 5.3: Detailed operational flowchart

5.4 Algorithm

The proposed system follows a structured multi-stage pipeline for real-time gaze-driven attention analysis. The complete algorithm consists of face detection, landmark extraction, head pose estimation, geometric gaze computation, screen mapping, fixation detection, and heatmap generation.

5.4.1 Face Detection and Landmark Extraction

A pretrained facial landmark detection model is used to detect the face region from the webcam stream. The model extracts dense facial landmarks, including eye contours, iris centers, nose tip, and facial boundary points.

These landmarks serve as reference points for head pose estimation and gaze vector computation.

5.4.2 Head Pose Estimation

Head pose is estimated using the Perspective-n-Point (PnP) algorithm. Given a set of known 3D facial reference points and their corresponding 2D image projections, the rotation and translation vectors are computed.

The transformation is represented as:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K[R \mid t] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \tag{5.1}$$

where:

- (X, Y, Z) are 3D facial model coordinates,
- (u, v) are image plane coordinates,
- K is the camera intrinsic matrix,
- R is the rotation matrix,
- t is the translation vector.

The resulting yaw, pitch, and roll angles represent head orientation.

5.4.3 Geometric Gaze Vector Estimation

Using the estimated head pose and eye center coordinates, a 3D gaze direction vector is computed geometrically.

The gaze vector is projected onto the 2D screen plane using a calibrated transformation matrix obtained during the calibration phase.

5.4.4 Calibration-Based Screen Mapping

A multi-point calibration procedure is performed before the session begins. Known screen coordinates are displayed, and predicted gaze coordinates are recorded.

A transformation function $f(x, y)$ is computed to minimize mapping error:

$$(x_{screen}, y_{screen}) = f(x_{pred}, y_{pred}) \tag{5.2}$$

This reduces systematic bias and improves spatial accuracy.

5.4.5 Media Synchronization

The system synchronizes gaze coordinates with the presented visual stimulus to ensure accurate attention mapping.

- **Dynamic Stimuli (Videos):** For each gaze sample, the current video timestamp is retrieved. For embedded videos, this is obtained via the platform's IFrame API. For local videos, the HTML5 video element's `currentTime` property is used.
- **Static Stimuli (Images/Photos):** Since images lack a native timeline, the system uses a high-resolution **Session Timer**. Every gaze point is recorded with an offset (in milliseconds) from the moment the image was first displayed. This allows for temporal analysis of how the user's attention explored the static content.

Each gaze record is stored with its temporal reference:

$$(x, y, t_{offset})$$

5.4.6 Fixation Detection (I-DT Algorithm)

The Dispersion-Threshold Identification (I-DT) algorithm is applied to detect fixations.

A fixation is identified when gaze points remain within a predefined spatial dispersion threshold D_{max} for a minimum duration T_{min}.

$$Dispersion = (x_{max} - x_{min}) + (y_{max} - y_{min}) \tag{5.3}$$

If:

$$Dispersion < D_{max} \quad \text{for duration} \geq T_{min}$$

the cluster is classified as a fixation.

5.4.7 Area of Interest (AOI) Classification

The screen is divided into predefined or user-defined regions known as Areas of Interest (AOIs). Each fixation is mapped to an AOI to compute attention distribution across different regions of the video frame.

5.4.8 Heatmap Generation

Accumulated gaze points are processed using Kernel Density Estimation (KDE) to generate a smooth attention heatmap:

$$H(x, y) = \sum_{i=1}^{N} K\left(\frac{(x - x_i)^2 + (y - y_i)^2}{h^2}\right) \tag{5.4}$$

where:

- (x_i, y_i) are gaze coordinates,
- K is the kernel function,
- h is the bandwidth parameter.

The resulting heatmap visually represents attention intensity.

5.4.9 Engagement Metric Computation

From detected fixations and AOI mapping, the system computes high-level engagement metrics for both video and image media. The core computation involves comparing gaze coordinates (x, y) with the bounding box coordinates of the Stimulus Area of Interest (AOI) $[X_{min}, Y_{min}, X_{max}, Y_{max}]$.

- **Focus Time (T_{focus}):** Cumulative time where gaze is within the stimulus boundaries.
- **Distraction Time ($T_{distract}$):** Cumulative time where gaze is outside the stimulus area or no face is detected.
- **Focus Score:** Percentage of active engagement calculated as:

$$Focus\ Score = \left(\frac{T_{focus}}{T_{total}}\right) \times 100\% \tag{5.5}$$

- **Scanpath Analysis (for Photos):** Sequential visualization of fixations and saccades to determine the order in which visual elements were attended to.
- **Heatmap Density**: Spatial distribution of attention over the media surface.

These metrics provide quantitative behavioral analysis beyond visual heatmap representation, allowing for objective assessment of content effectiveness across dynamic and static media.

5.5 Accuracy Evaluation

To evaluate the performance of the proposed gaze tracking system, multiple quantitative metrics are used to assess spatial precision, classification reliability, and system responsiveness.

5.5.1 Calibration-Based Spatial Accuracy

Spatial accuracy is measured using controlled calibration points displayed on the screen. For each calibration point, the true screen coordinate (x_{true}, y_{true}) is known, and the system predicts a gaze coordinate (x_{pred}, y_{pred}).

The Euclidean pixel error is calculated as:

$$Error = \sqrt{(x_{true} - x_{pred})^2 + (y_{true} - y_{pred})^2} \tag{5.6}$$

The Mean Calibration Error (MCE) across N calibration points is computed as:

$$MCE = \frac{1}{N} \sum_{i=1}^{N} Error_i \tag{5.7}$$

Lower mean error indicates higher spatial accuracy.

5.5.2 Region-Based Accuracy

The screen is divided into predefined regions (e.g., 4 or 9 regions). A prediction is considered correct if the estimated gaze falls within the intended region.

$$Region\ Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} \times 100\% \tag{5.8}$$

This metric evaluates the practical usability of the system in attention analysis.

5.5.3 Fixation Stability

Fixation stability is measured by computing the variance of gaze points within a detected fixation cluster:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})^2 \tag{5.9}$$

Lower variance indicates stable fixation detection.

5.5.4 System Latency

System latency is defined as the time delay between frame capture and gaze coordinate output. Average latency is measured in milliseconds (ms) to evaluate real-time performance capability.

5.5.5 Session Consistency

To assess repeatability, multiple sessions are conducted under similar conditions. Correlation analysis is performed between sessions to evaluate consistency of attention patterns.

These evaluation metrics ensure that the proposed system is validated for behavioral attention analysis rather than medical-grade precision.

5.6 Technology Stack

The system is developed utilizing a robust mix of contemporary web and computer vision technologies, distributed across frontend interfaces, processing analytics, and visualization tools.

Table 5.1: Technology Stack Breakdown

Category	Tool/Technology	Purpose
Frontend	HTML, CSS, JavaScript	Core web interface development and media player construction.
Video Integration	Embedded Video APIs (e.g., YouTube IFrame API)	Video embedding, seamless playback control, and timestamp synchronization.
Vision Frameworks	MediaPipe Face Mesh, OpenCV	High-fidelity facial and eye landmark extraction; general image frame processing.
Processing & Analytics	Python, NumPy	Mathematical computation for geometric gaze mapping, Fixation Detection (I-DT), and engagement scaling.
Data Visualization	Heatmap.js, Chart.js, D3.js	Generation of continuous KDE attention heatmaps and dynamic temporal engagement timelines.
Hardware	Standard RGB Webcam, Personal Computer	Image capture and local processing without specialized infrared optical sensors.

6. EXPECTED OUTPUT

The proposed system is expected to produce the following outcomes:

- A functional webcam-based gaze tracking application capable of operating in real time during video playback.
- Accurate mapping of user gaze coordinates onto video frames after calibration.
- Detection of visual fixations using dispersion-based algorithms.
- Generation of visual attention heatmaps representing gaze intensity distribution over video content.
- Computation of quantitative engagement and reliability metrics such as:
  - **Focus Score**: Overall percentage of session engagement.
  - **Average Confidence Score**: Mean tracking reliability during the session.
  - **Distraction Index**: Breakdown of distracted intervals.
  - **Scanpath Clarity (for static images)**: Path analysis of visual exploration.
  - Total fixation duration
  - Fixation count per region
  - Attention percentage per Stimulus Area of Interest (AOI)
  - Temporal attention variation across the playback or viewing timeline
- A comprehensive **Session Report Dashboard** that summarizes accuracy, confidence, and engagement metrics in a unified visual interface.
- Exportable session data in structured format (e.g., CSV) for further analysis.
- Performance evaluation results including calibration error, region-based accuracy, and system latency measurements.
- A web-based user interface with embedded video playback that allows users to paste any public video URL for attention analysis.

The expected attention heatmap output will visually represent gaze intensity distribution using a color-coded overlay on each video frame, as illustrated in the experimental mockup in Figure 6.1. Warmer colors (red, orange) indicate regions of high visual

attention, while cooler colors (blue, green) indicate lower attention. The attention timeline will display temporal engagement variation across the video duration, enabling identification of peak-attention and drop-off segments.

Figure 1: Attention Heatmap Overlay on Video Content.

Figure 6.1: Experimental mockup of an attention heatmap overlay

The final deliverable will be a complete prototype system demonstrating real-time gaze-driven attention analysis integrated with embedded video playback, accompanied by quantitative evaluation results and exportable analysis reports.

7. APPLICATIONS

The proposed gaze-driven visual attention analysis system has practical applications across multiple domains:

- **Educational Video Analysis:** Identifying which segments of recorded lectures attract student attention and detecting attention drop-off points to improve content delivery and instructional design.
- **Advertisement and Trailer Evaluation:** Assessing which scenes and visual elements in advertisements or trailers receive maximum viewer focus, aiding content optimization and creative decision-making.
- **UI/UX and Usability Research:** Studying how users visually interact with both live websites (via screenshots) and video interfaces to improve layout, visual hierarchy, and overall user experience.
- **Print and Digital Advertisement Analysis:** Assessing which segments of static advertisements or video commercials receive maximum viewer focus, aiding creative decision-making.
- **Human–Computer Interaction (HCI) Research:** Providing gaze-based behavioral data for understanding visual behavior during multimedia interaction and screen-based tasks.
- **Market Research and A/B Testing:** Comparing attention patterns across different versions of video content to support data-driven marketing and creative decisions.

8. TIME SCHEDULE (GANTT CHART)

The planned timeline for the research, development, and evaluation phases of the project is detailed in the Gantt Chart shown in Figure 8.1.

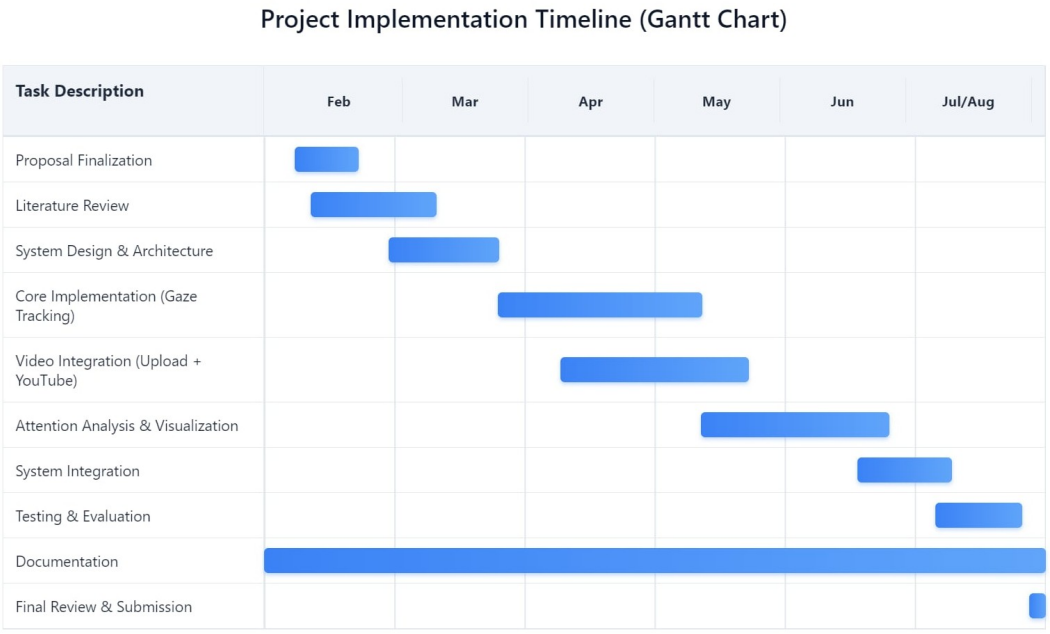

Figure 8.1: Gantt Chart

9. COST ESTIMATION

The estimated financial requirements for the proposed system are summarized in Table 9.1.

Table 9.1: Estimated Project Cost

Item	**Estimated Cost**
Webcam	Existing Resource
Laptop / PC	Existing Resource
Software Tools (OpenCV, MediaPipe, Python)	Free / Open Source
Internet Usage	Existing Facility
Printing and Documentation	NPR 500
Miscellaneous Expenses	NPR 500
Total Estimated Cost	**NPR 1000**

10. FUTURE SCOPE

The current system provides a functional foundation for gaze-driven attention analysis. Several enhancements can be explored in future iterations:

- **Deep Learning-Based Gaze Estimation:** Integrating CNN-based gaze prediction models trained on large-scale datasets such as MPIIGaze to improve spatial accuracy beyond geometric methods.
- **Multi-User Aggregated Analysis:** Extending the system to aggregate attention data from multiple users to generate comparative and population-level attention insights.
- **Adaptive Recalibration:** Implementing continuous calibration adjustment during sessions to compensate for head movement and posture changes over time.
- **Automatic AOI Detection:** Using object detection models to automatically identify and define Areas of Interest within video frames rather than relying on predefined screen regions.
- **Learning Management System Integration:** Integrating with platforms such as Moodle or Google Classroom to provide attention analytics for online education environments.
- **Browser Extension Deployment:** Packaging the system as a browser extension for broader accessibility and easier deployment across different video platforms.

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